



Deep learning-based human gunshot wounds classification

Renato Queiroz Nogueira Lira¹ · Luana Geovana Motta de Sousa¹ · Maisa Luana Memoria Pinho¹ · Renan Cesar Pinto da Silva Andrade de Lima¹ · Pedro Garcia Freitas² · Bruno Scholles Soares Dias³ · Andreia Cristina Breda de Souza⁴ · André Ferreira Leite¹

Received: 10 June 2024 / Accepted: 18 October 2024

© The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2024

Abstract

In this paper, we present a forensic perspective on classifying gunshot wound patterns using Deep Learning (DL). Although DL has revolutionized various medical specialties, such as automating tasks like medical image classification, its applications in forensic contexts have been limited despite the inherently visual nature of the field. This study investigates the application of DL techniques (59 architectures) to classify gunshot wounds in a forensic context, focusing on distinguishing between entry and exit wounds and determining the Medical-Legal Shooting Distance (MLSD), which classifies wounds as contact, close range, or distant, based on digital images from real crime scene cases. A comprehensive database was constructed with 2,551 images, including 1,883 entries and 668 exit wounds. The ResNet152 architecture demonstrated superior performance in both entry and exit wound classification and MLSD categorization. For the first task, achieved accuracy of 86.90% and an AUC of 82.09%. For MLSD, the ResNet152 showed an accuracy of 92.48% and AUC up to 94.36%, though sample imbalance affected the metrics. Our findings underscore the challenges of standardizing wound images due to varying capture conditions but reflect the practical realities of forensic work. This research highlights the significant potential of DL in enhancing forensic pathology practices, advocating for Artificial Intelligence (AI) as a supportive tool to complement human expertise in forensic investigations.

Keywords Forensic medicine · Gunshot wounds · Artificial intelligence · Deep learning

Introduction

Firearms correspond to one of the most used instruments worldwide for committing homicides, being related to 75% of cases in the Americas in 2021 [1]. In Brazil, the country with the highest number of firearm-related fatalities, there are more than 47,000 deaths attributed to firearms annually [2]. Given these high rates, it is common in forensic fieldwork to encounter gunshot wounds during external body examinations, whether at the crime scene or during an autopsy. When these are found, among various procedures, it is necessary to distinguish between entry and exit wounds and to classify the entry wounds according to the distance from which the shot was fired [3, 4].

Some characteristics, such as shape, size, and margins, allow for relatively easy differentiation between entry and exit wounds [5]. However, these characteristics are not always clearly visible, or they may be partially absent or even overlapping, complicating the accurate classification. Additionally, other factors can influence wound production, including the caliber and type of ammunition, the angle and speed of the shot, barriers encountered by the projectile, and the region of the body impacted [6, 7].

Similarly, the classification of the shot distance can be influenced by the same aforementioned factors [6]. This distance has a medico-legal classification, which is based on the characteristics presented by the injury itself and its naked eye appearances. These categories include: distant (or medium-distance) wound, close (or near) range wound and contact wound [7, 8]. Some authors also introduce additional categories between close-range and contact, such as near-contact and loose-contact [6]. Moreover, it is important to emphasize that this is a medico-legal classification based on the

Luana Geovana Motta de Sousa, Maisa Luana Memoria Pinho, and Renan Cesar Pinto da Silva Andrade de Lima contributed equally to this work.

Extended author information available on the last page of the article

traces found on the victim, not necessarily reflecting the actual event, and serves as an indication or suggestion of what occurred at the crime scene [7].

Regardless, for forensic investigation purposes, any additional information can be crucial in aiding the elucidation of what happened, how it happened, and who was involved. However, to ensure impartiality and justice, it is important that the evidence produced is based on and grounded in scientific criteria [9].

In turn, in recent years, there has been an increase in the research and perspectives of use of AI with Machine Learning (ML) across various sectors, including forensic medicine [10–15]. In the pursuit of faster and more robust crime resolution, image recognition and classification have seen significant advancements, including initiating studies on the classification of firearm injuries [16, 17]. Despite these developments contributing to the enhancement of forensic methodologies and the efficiency of criminal investigations, and being the studies that initiated research in the area, they involved the use of only a few images and were based on injuries in pig carcasses [17] or, when trained with autopsy photographs of wounds on human bodies, did not analyze the MLSD aspect [16].

The current study aimed to develop a ML model that can analyze and classify photographs of real crime scene forensic cases of gunshot wounds in relation to determining whether they are entry or exit wounds and to classify MLSD of the gunshot wounds.

Material and methods

Figure 1 illustrated the framework employed during the development of this work. It includes three main stages responsible for preprocessing the captured photographs from the crime scene, classifying the preprocessed images using a deep neural network, and finally estimating the classification according to a specific forensic task. The following subsections details each stage.

Data acquisition

In this study, photographic records of homicide and suicide cases examined by crime scene forensic experts and reports by medical examiners from incidents attended by the Federal District Civil Police (PCDF) (Brasília, Federal District, Brazil) from the January of 2012 until March of 2022 were used. From the selected time period, a total of 2,012,584 photographs were stored. This dataset was obtained with the proper authorization of PCDF and the Committee for Ethics in Research of the University of Brasília approved this research (Protocol number 54418221.9.0000.0030).

The images and relevant data related to the wounds, such as age, sex, anatomical location, distance, and wound category, the following inclusion criteria were established:

1. the victim was examined and photographed in the crime scene by forensic experts;
2. the body did not show advanced signs of decomposition (e.g., greenish discoloration of the skin of the anterior abdominal wall in the right iliac fossa region);
3. the photos from the scene examination were in focus and with sufficient visualization of the wounds;
4. the victim was identified, due to the need for additional data - sex and age at the time of death;
5. there were bullets stored in the bodies;
6. the bullets had been recovered during the autopsy at the Legal Medical Institute (IML/PCDF);
7. the bullets recovered at the IML had been sent to the Institute of Criminalistics (IC/PCDF);
8. it was possible to determine the caliber(s) of the bullet(s) at the Forensic Ballistics Section (SBF/IC/PCDF);
9. only one caliber involved per victim was determined at SBF;
10. the respective autopsy reports, made by experienced forensics pathologists, indicates which anatomical location, shot distance (distant, close or contact) and category (entry or exit) the photographed wound corresponds to. These reports correspond to the gold standard.

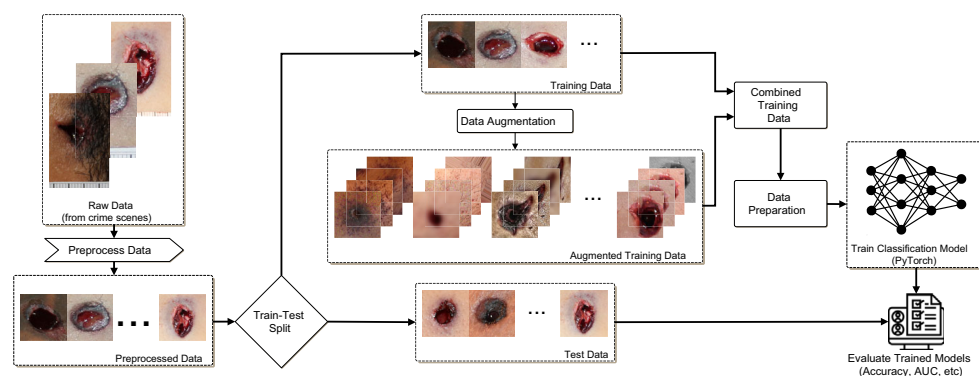


Fig. 1 Developed framework for data augmentation, classification modeling, and used to evaluate the proposed methodology

Table 1 Cases and images per sex and wound type (entry or exit)

Sex	# Cases (%)	# Images (%)	# Entries (%)	# Exits (%)
Female	24 (4.91)	114 (4.47)	85 (4.51)	29 (4.34)
Male	465 (95.09)	2437 (95.53)	1798 (95.49)	639 (95.66)
Total	489 (100)	2551 (100)	1883 (100)	668 (100)

These criteria were designed based on the working protocol guidelines of the PCDF and, due to the use of this database, several wounds and cases were excluded from this study, reducing the available sample. However, these data may be used for further analyses, potentially involving caliber as one of the factors to be studied.

Data preprocessing

The images retrieved from these cases were then manually cropped to isolate the wounds in a square format. This manual processing was handled using the GNU Image Manipulation Program (GIMP) image processing software, resulting in a preprocessed dataset of 2,551 images of gunshot wounds. These preprocessed images were then categorized according to wound type and MLSD. The wound types were categorized into entry and exit wounds, corresponding to 1,883 entry and 668 exit wounds. Table 1 details the wound types according to the number of cases, associated images, and sex. Moreover, considering only the subset of entry wound images, we obtained 1,609 from distance shots, 232 from close range, and 42 from contact shots as per the MLSD categorization, as depicted in Table 2.

As observed in Tables 1, 2 and 3, the dataset under study exhibits a significant imbalance. Such imbalance is typical in forensic datasets, where certain types of wounds or shooting distances are often underrepresented compared to others [18]. This imbalance stems from real-world crime cases, where certain types of wounds are rare or sensitive, leading to limited sample availability. In response to this, techniques were applied during data preprocessing when constructing the training pipeline for the models under investigation.

Learning-based classification

Considering the curated data as described in the previous section, we identified that the recognition of forensic characteristics of gunshot wound images can be expressed as an

Table 2 Entry wounds images per medico-legal distance classification

Distance	# Images (%)
Distant	1609 (85.45)
Close range	232 (12.32)
Contact	42 (2.23)
Total	1883 (100)

image classification problem. More specifically, the identification of the wound type is a binary classification problem (i.e., entry or exit), while the categorization of the MLSD is a multiclass classification problem, as can be inferred from Table 2. In other words, both are Computer Vision (CV) problems. Therefore, this study tested the potential of a series of DL models established in the literature for CV problems.

The neural networks used in this study were achieved from TorchVision [19], popular library that is part of the PyTorch [20] ecosystem. TorchVision provides tools and utilities specifically designed for computer vision tasks, such as state-of-the-art architectures and its pretrained models. From this library, fifty-nine state-of-the-art DL architectures were utilized to analyze forensic images for the differentiation of entry and exit wounds as well as the determination of MLSD. These models are listed in Tables 4 and 5. Additionally, we employed a supplementary abstraction software layer using Skorch [21], a library that provides a high-level interface for working with PyTorch models in a scikit-learn-like way. This additional layer was included to add Scikit-Learn [22] compatibility and make it easy to use PyTorch models with the scikit-learn ecosystem. This allows our software implementation to leverage tools and utilities from Scikit-Learn, such as grid search, cross-validation, hyperparameter tuning, and pipelines, with PyTorch models. Thus, with the use of Skorch, our implementation benefits from several desirable properties in terms of software engineering, such as decoupling, maintainability, efficiency, correctness, reusability, testability, etc. Our concern on these properties was to ensure reliability in conducting our experiments.

As can be noticed from Tables 1, 2 and 3, the studied dataset is highly imbalanced. This imbalance is usually common in forensic datasets, where certain types of wounds or shooting distances may be underrepresented compared to others. The data imbalance emerges from the real-world crime cases, where some type of wounds can often be lim-

Table 3 Number of wound images per anatomical location on the body

Anatomical location	# Images (%)
Head and neck	548 (21.48)
Trunk	1310 (51.35)
Upper limbs	503 (19.72)
Lower limbs	190 (7.45)
Total	2551 (100)

Table 4 Performance metrics for different neural network architectures using hold-out validation to predict the wound type

Architecture	Variant	ACC	Precision			Recall			F1-Score			AUC			Specificity		
			M	m	W	M	m	W	M	m	W	M	m	W	M	m	W
Alexnet [28]		0.765	0.695	0.765	0.780	0.716	0.765	0.765	0.703	0.765	0.771	0.716	0.716	0.716	0.716	0.765	0.765
DenseNet [29]	121	0.736	0.640	0.736	0.727	0.629	0.736	0.736	0.634	0.736	0.731	0.629	0.629	0.629	0.629	0.736	0.736
	161	0.753	0.659	0.753	0.737	0.635	0.753	0.753	0.644	0.753	0.742	0.635	0.635	0.635	0.635	0.753	0.753
	169	0.777	0.700	0.777	0.768	0.679	0.777	0.777	0.688	0.777	0.772	0.679	0.679	0.679	0.679	0.777	0.777
	201	0.744	0.663	0.744	0.751	0.671	0.744	0.744	0.667	0.744	0.747	0.671	0.671	0.671	0.671	0.744	0.744
EfficientNet [30, 31]	B0	0.744	0.646	0.744	0.727	0.624	0.744	0.744	0.632	0.744	0.734	0.624	0.624	0.624	0.624	0.744	0.744
	B1	0.734	0.627	0.734	0.713	0.603	0.734	0.734	0.610	0.734	0.720	0.603	0.603	0.603	0.603	0.734	0.734
	B2	0.750	0.656	0.750	0.734	0.631	0.750	0.750	0.639	0.750	0.740	0.631	0.631	0.631	0.631	0.750	0.750
	B3	0.757	0.659	0.757	0.728	0.604	0.757	0.757	0.615	0.757	0.732	0.604	0.604	0.604	0.604	0.757	0.757
	B4	0.715	0.588	0.715	0.684	0.566	0.715	0.715	0.569	0.715	0.695	0.566	0.566	0.566	0.566	0.715	0.715
	B5	0.744	0.608	0.744	0.686	0.535	0.744	0.744	0.517	0.744	0.683	0.535	0.535	0.535	0.535	0.744	0.744
	B6	0.759	0.667	0.759	0.740	0.633	0.759	0.759	0.644	0.759	0.745	0.633	0.633	0.633	0.633	0.759	0.759
	B7	0.750	0.375	0.750	0.563	0.500	0.750	0.750	0.429	0.750	0.644	0.500	0.500	0.500	0.500	0.750	0.750
	V2L	0.782	0.710	0.782	0.760	0.638	0.782	0.782	0.654	0.782	0.759	0.638	0.638	0.638	0.638	0.782	0.782
	V2M	0.726	0.602	0.726	0.693	0.572	0.726	0.726	0.577	0.726	0.703	0.572	0.572	0.572	0.572	0.726	0.726
	V2S	0.721	0.621	0.721	0.714	0.614	0.721	0.721	0.617	0.721	0.717	0.614	0.614	0.614	0.614	0.721	0.721
MaxVit [32]		0.780	0.703	0.780	0.761	0.653	0.780	0.780	0.668	0.780	0.764	0.653	0.653	0.653	0.653	0.780	0.780
MNASNet [33]	5	0.705	0.589	0.705	0.688	0.578	0.705	0.705	0.582	0.705	0.695	0.578	0.578	0.578	0.578	0.705	0.705
	75	0.717	0.590	0.717	0.686	0.567	0.717	0.717	0.571	0.717	0.696	0.567	0.567	0.567	0.567	0.717	0.717
	10	0.682	0.570	0.682	0.677	0.568	0.682	0.682	0.569	0.682	0.680	0.568	0.568	0.568	0.568	0.682	0.682
	13	0.746	0.659	0.746	0.744	0.656	0.746	0.746	0.657	0.746	0.745	0.656	0.656	0.656	0.656	0.746	0.746
MobileNet [34, 35]	V2	0.719	0.609	0.719	0.702	0.593	0.719	0.719	0.598	0.719	0.709	0.593	0.593	0.593	0.593	0.719	0.719
	V3Large	0.690	0.566	0.690	0.672	0.557	0.690	0.690	0.559	0.690	0.680	0.557	0.557	0.557	0.557	0.690	0.690
	V3Small	0.703	0.585	0.703	0.685	0.574	0.703	0.703	0.577	0.703	0.693	0.574	0.574	0.574	0.574	0.703	0.703
Regnet [36]	X16GF	0.690	0.534	0.690	0.648	0.524	0.690	0.690	0.521	0.690	0.664	0.524	0.524	0.524	0.524	0.690	0.690
	X32GF	0.724	0.580	0.724	0.675	0.546	0.724	0.724	0.544	0.724	0.687	0.546	0.546	0.546	0.546	0.724	0.724
	X400MF	0.663	0.528	0.663	0.646	0.525	0.663	0.663	0.526	0.663	0.654	0.525	0.525	0.525	0.525	0.663	0.663
	X800MF	0.703	0.563	0.703	0.667	0.546	0.703	0.703	0.547	0.703	0.680	0.546	0.546	0.546	0.546	0.703	0.703
	X8GF	0.699	0.558	0.699	0.664	0.543	0.699	0.699	0.544	0.699	0.677	0.543	0.543	0.543	0.543	0.699	0.699
	Y16GF	0.711	0.571	0.711	0.672	0.549	0.711	0.711	0.549	0.711	0.685	0.549	0.549	0.549	0.549	0.711	0.711

Table 4 continued

Architecture	Variant	ACC	Precision			Recall			F1-Score			AUC			Specificity		
			M	m	W	M	m	W	M	m	W	M	m	W	M	m	W
ResNet [37]	Y32GF	0.671	0.533	0.671	0.648	0.528	0.671	0.671	0.528	0.671	0.659	0.528	0.528	0.528	0.528	0.671	0.671
	Y400MF	0.674	0.540	0.674	0.654	0.535	0.674	0.674	0.536	0.674	0.663	0.535	0.535	0.535	0.535	0.674	0.674
	Y800MF	0.703	0.565	0.703	0.669	0.549	0.703	0.703	0.550	0.703	0.681	0.549	0.549	0.549	0.549	0.703	0.703
	Y8GF	0.703	0.563	0.703	0.667	0.546	0.703	0.703	0.547	0.703	0.680	0.546	0.546	0.546	0.546	0.703	0.703
	152	0.869	0.827	0.869	0.868	0.821	0.869	0.869	0.824	0.869	0.869	0.821	0.821	0.821	0.821	0.869	0.869
	101	0.863	0.822	0.863	0.860	0.803	0.863	0.863	0.811	0.863	0.861	0.803	0.803	0.803	0.803	0.863	0.863
	50	0.850	0.802	0.850	0.848	0.792	0.850	0.850	0.797	0.850	0.849	0.792	0.792	0.792	0.792	0.850	0.850
	34	0.850	0.799	0.850	0.852	0.806	0.850	0.850	0.802	0.850	0.851	0.806	0.806	0.806	0.806	0.850	0.850
	18	0.817	0.758	0.817	0.810	0.734	0.817	0.817	0.744	0.817	0.812	0.734	0.734	0.734	0.734	0.817	0.817
	ShuffleNet [38]	V2x05	0.730	0.628	0.730	0.717	0.614	0.730	0.730	0.619	0.730	0.722	0.614	0.614	0.614	0.730	0.730
SqueezeNet [39]	V2x10	0.701	0.533	0.701	0.646	0.520	0.701	0.701	0.512	0.701	0.664	0.520	0.520	0.520	0.520	0.701	0.701
	V2x15	0.711	0.580	0.711	0.679	0.560	0.711	0.711	0.563	0.711	0.690	0.560	0.560	0.560	0.560	0.711	0.711
	V2x20	0.692	0.551	0.692	0.660	0.539	0.692	0.692	0.540	0.692	0.672	0.539	0.539	0.539	0.539	0.692	0.692
	10	0.750	0.375	0.750	0.563	0.500	0.750	0.750	0.429	0.750	0.644	0.500	0.500	0.500	0.500	0.750	0.750
SwinT [40]	11	0.750	0.375	0.750	0.563	0.500	0.750	0.750	0.429	0.750	0.644	0.500	0.500	0.500	0.500	0.750	0.750
	B	0.728	0.614	0.728	0.703	0.591	0.728	0.728	0.597	0.728	0.712	0.591	0.591	0.591	0.591	0.728	0.728
	S	0.671	0.536	0.671	0.650	0.531	0.671	0.671	0.531	0.671	0.660	0.531	0.531	0.531	0.531	0.671	0.671
	T	0.713	0.594	0.713	0.691	0.578	0.713	0.713	0.583	0.713	0.700	0.578	0.578	0.578	0.578	0.713	0.713
VGG [41]	V2B	0.750	0.375	0.750	0.563	0.500	0.750	0.750	0.429	0.750	0.644	0.500	0.500	0.500	0.500	0.750	0.750
	V2S	0.713	0.557	0.713	0.661	0.533	0.713	0.713	0.528	0.713	0.676	0.533	0.533	0.533	0.533	0.713	0.713
	V2T	0.750	0.375	0.750	0.563	0.500	0.750	0.750	0.429	0.750	0.644	0.500	0.500	0.500	0.500	0.750	0.750
	11	0.763	0.673	0.763	0.742	0.633	0.763	0.763	0.645	0.763	0.748	0.633	0.633	0.633	0.633	0.763	0.763
ViT [42]	13	0.790	0.720	0.790	0.774	0.671	0.790	0.790	0.687	0.790	0.777	0.671	0.671	0.671	0.671	0.790	0.790
	16	0.782	0.706	0.782	0.772	0.685	0.782	0.782	0.694	0.782	0.776	0.685	0.685	0.685	0.685	0.782	0.782
	19	0.777	0.700	0.777	0.768	0.679	0.777	0.777	0.688	0.777	0.772	0.679	0.679	0.679	0.679	0.777	0.777
	B16	0.713	0.587	0.713	0.684	0.567	0.713	0.713	0.571	0.713	0.695	0.567	0.567	0.567	0.567	0.713	0.713
ViT [42]	B32	0.674	0.559	0.674	0.669	0.557	0.674	0.674	0.558	0.674	0.671	0.557	0.557	0.557	0.557	0.674	0.674
	L16	0.678	0.544	0.678	0.656	0.538	0.678	0.678	0.539	0.678	0.666	0.538	0.538	0.538	0.538	0.678	0.678
	L32	0.692	0.563	0.692	0.669	0.553	0.692	0.692	0.555	0.692	0.679	0.553	0.553	0.553	0.553	0.692	0.692

Considering each metric separately, the best results in terms of maximum absolute value are **boldfaced**. Symbols M, m, and W illustrates macro, micro, and weighted, respectively

Table 5 Performance metrics for different neural network architectures using hold-out validation to predict the medico-legal shooting distance

Architecture	Variant	ACC	Precision			Recall			F1-Score			AUC			Specificity		
			M	m	W	M	m	W	M	m	W	M	m	W	M	m	W
Alexnet [28]		0.889	0.537	0.889	0.862	0.378	0.889	0.889	0.395	0.889	0.856	0.541	0.916	0.555	0.703	0.944	0.222
DenseNet [29]	121	0.877	0.422	0.877	0.825	0.356	0.877	0.877	0.359	0.877	0.841	0.521	0.908	0.527	0.685	0.939	0.177
	161	0.894	0.824	0.894	0.871	0.465	0.894	0.894	0.528	0.894	0.863	0.589	0.921	0.569	0.712	0.947	0.243
	169	0.883	0.467	0.883	0.820	0.388	0.883	0.883	0.398	0.883	0.844	0.548	0.912	0.563	0.708	0.942	0.242
	201	0.902	0.634	0.902	0.883	0.482	0.902	0.902	0.527	0.902	0.886	0.627	0.927	0.659	0.772	0.951	0.415
EfficientNet [30, 31]	B0	0.877	0.758	0.877	0.857	0.438	0.877	0.877	0.483	0.877	0.858	0.583	0.908	0.591	0.728	0.939	0.305
	B1	0.886	0.670	0.886	0.852	0.462	0.886	0.886	0.508	0.886	0.857	0.585	0.914	0.564	0.709	0.943	0.242
	B2	0.864	0.393	0.864	0.817	0.360	0.864	0.864	0.364	0.864	0.836	0.523	0.898	0.531	0.687	0.932	0.198
	B3	0.900	0.560	0.900	0.873	0.400	0.900	0.900	0.424	0.900	0.869	0.561	0.925	0.582	0.722	0.950	0.265
	B4	0.883	0.450	0.883	0.837	0.376	0.883	0.883	0.386	0.883	0.851	0.539	0.912	0.552	0.701	0.942	0.221
	B5	0.869	0.357	0.869	0.806	0.345	0.869	0.869	0.340	0.869	0.833	0.513	0.902	0.523	0.667	0.943	0.114
	B6	0.886	0.295	0.886	0.785	0.333	0.886	0.886	0.313	0.886	0.832	0.500	0.914	0.500	0.667	0.943	0.114
	B7	0.886	0.463	0.886	0.834	0.342	0.886	0.886	0.332	0.886	0.837	0.508	0.914	0.511	0.674	0.943	0.136
	V2L	0.880	0.617	0.880	0.854	0.448	0.880	0.880	0.488	0.880	0.861	0.588	0.910	0.593	0.729	0.940	0.306
	V2M	0.883	0.614	0.883	0.846	0.414	0.883	0.883	0.448	0.883	0.852	0.558	0.912	0.552	0.701	0.942	0.221
MaxVit [32]	V2S	0.875	0.592	0.875	0.843	0.428	0.875	0.875	0.465	0.875	0.852	0.570	0.906	0.569	0.712	0.937	0.263
		0.880	0.765	0.880	0.859	0.486	0.880	0.880	0.549	0.880	0.862	0.607	0.910	0.593	0.729	0.940	0.305
MNASNet [33]	5	0.844	0.344	0.844	0.806	0.344	0.844	0.844	0.342	0.844	0.824	0.519	0.883	0.541	0.694	0.922	0.238
	75	0.872	0.521	0.872	0.843	0.436	0.872	0.872	0.462	0.872	0.853	0.577	0.904	0.578	0.719	0.936	0.284
	10	0.872	0.566	0.872	0.837	0.418	0.872	0.872	0.451	0.872	0.850	0.568	0.904	0.578	0.719	0.936	0.283
	13	0.869	0.409	0.869	0.824	0.371	0.869	0.869	0.378	0.869	0.842	0.533	0.902	0.544	0.696	0.934	0.219
MobileNet [34, 35]	V2	0.886	0.424	0.886	0.831	0.360	0.886	0.886	0.361	0.886	0.848	0.531	0.914	0.553	0.702	0.943	0.221
	V3Large	0.872	0.411	0.872	0.829	0.381	0.872	0.872	0.388	0.872	0.847	0.546	0.904	0.567	0.711	0.936	0.262
	V3Small	0.891	0.548	0.891	0.862	0.362	0.891	0.891	0.367	0.891	0.849	0.526	0.918	0.535	0.690	0.946	0.179
Regnet [36]	X16GF	0.855	0.294	0.855	0.781	0.322	0.855	0.855	0.307	0.855	0.817	0.489	0.891	0.483	0.655	0.928	0.111
	X32GF	0.872	0.370	0.872	0.808	0.346	0.872	0.872	0.341	0.872	0.834	0.511	0.904	0.514	0.676	0.936	0.155
	X400MF	0.844	0.358	0.844	0.808	0.353	0.844	0.844	0.353	0.844	0.825	0.520	0.883	0.530	0.687	0.922	0.217
	X800MF	0.855	0.487	0.855	0.804	0.377	0.855	0.855	0.396	0.855	0.826	0.527	0.891	0.515	0.677	0.928	0.175
	X8GF	0.850	0.349	0.850	0.802	0.346	0.850	0.850	0.344	0.850	0.824	0.510	0.887	0.512	0.675	0.925	0.174
	Y16GF	0.875	0.398	0.875	0.820	0.355	0.875	0.875	0.357	0.875	0.840	0.523	0.906	0.537	0.691	0.937	0.199

Table 5 continued

Architecture	Variant	ACC	Precision			Recall			F1-Score			AUC			Specificity		
			M	m	W	M	m	W	M	m	W	M	m	W	M	m	W
ResNet [37]	Y32GF	0.864	0.659	0.864	0.814	0.380	0.864	0.864	0.407	0.864	0.830	0.526	0.898	0.509	0.673	0.932	0.154
	Y400MF	0.869	0.333	0.869	0.798	0.336	0.869	0.869	0.326	0.869	0.829	0.505	0.902	0.512	0.675	0.934	0.155
	Y800MF	0.880	0.531	0.880	0.819	0.387	0.880	0.880	0.404	0.880	0.840	0.536	0.910	0.529	0.686	0.940	0.178
	Y8GF	0.861	0.324	0.861	0.794	0.333	0.861	0.861	0.323	0.861	0.825	0.502	0.895	0.507	0.672	0.930	0.154
	152	0.925	0.810	0.925	0.917	0.637	0.925	0.925	0.701	0.925	0.917	0.733	0.944	0.745	0.830	0.962	0.566
	101	0.914	0.813	0.914	0.903	0.571	0.914	0.914	0.642	0.914	0.896	0.674	0.935	0.665	0.776	0.957	0.416
	50	0.911	0.862	0.911	0.898	0.617	0.911	0.911	0.696	0.911	0.896	0.696	0.933	0.663	0.775	0.955	0.415
	34	0.897	0.732	0.897	0.885	0.591	0.897	0.897	0.643	0.897	0.889	0.691	0.923	0.687	0.792	0.949	0.478
ShuffleNet [38]	18	0.914	0.557	0.914	0.887	0.449	0.914	0.914	0.479	0.914	0.892	0.613	0.935	0.665	0.776	0.957	0.415
	V2x05	0.855	0.355	0.855	0.804	0.348	0.855	0.855	0.346	0.855	0.827	0.512	0.891	0.515	0.677	0.928	0.175
	V2x10	0.877	0.411	0.877	0.830	0.374	0.877	0.877	0.380	0.877	0.849	0.544	0.908	0.570	0.713	0.939	0.263
	V2x15	0.869	0.337	0.869	0.797	0.336	0.869	0.869	0.326	0.869	0.828	0.502	0.902	0.501	0.668	0.934	0.134
SqueezeNet [39]	V2x20	0.880	0.379	0.880	0.810	0.340	0.880	0.880	0.330	0.880	0.834	0.506	0.910	0.508	0.672	0.940	0.135
	10	0.095	0.032	0.095	0.009	0.333	0.095	0.095	0.058	0.095	0.016	0.500	0.321	0.500	0.667	0.547	0.905
SwinT [40]	11	0.886	0.295	0.886	0.785	0.333	0.886	0.886	0.313	0.886	0.832	0.500	0.914	0.500	0.667	0.943	0.114
	B	0.861	0.354	0.861	0.809	0.341	0.861	0.861	0.339	0.861	0.831	0.517	0.895	0.540	0.693	0.930	0.219
	S	0.855	0.522	0.855	0.816	0.395	0.855	0.855	0.420	0.855	0.831	0.539	0.891	0.526	0.684	0.928	0.197
	T	0.872	0.727	0.872	0.840	0.410	0.872	0.872	0.449	0.872	0.845	0.553	0.904	0.546	0.697	0.936	0.219
	V2B	0.886	0.295	0.886	0.785	0.333	0.886	0.886	0.313	0.886	0.832	0.500	0.914	0.500	0.667	0.943	0.114
VGG [41]	V2S	0.861	0.405	0.861	0.826	0.376	0.861	0.861	0.384	0.861	0.841	0.542	0.895	0.561	0.707	0.930	0.262
	V2T	0.872	0.419	0.872	0.826	0.363	0.872	0.872	0.369	0.872	0.841	0.527	0.904	0.535	0.690	0.936	0.199
	11	0.905	0.666	0.905	0.889	0.466	0.905	0.905	0.514	0.905	0.884	0.613	0.929	0.639	0.759	0.953	0.372
	13	0.875	0.542	0.875	0.845	0.437	0.875	0.875	0.467	0.875	0.856	0.582	0.906	0.590	0.727	0.937	0.305
ViT [42]	16	0.864	0.392	0.864	0.825	0.378	0.864	0.864	0.381	0.864	0.843	0.546	0.898	0.573	0.715	0.932	0.282
	19	0.883	0.578	0.883	0.852	0.431	0.883	0.883	0.466	0.883	0.858	0.574	0.912	0.573	0.716	0.942	0.264
	B16	0.838	0.342	0.838	0.801	0.342	0.838	0.838	0.340	0.838	0.819	0.510	0.879	0.517	0.678	0.919	0.195
	B32	0.875	0.425	0.875	0.837	0.390	0.875	0.875	0.400	0.875	0.853	0.558	0.906	0.590	0.727	0.937	0.305
	L16	0.869	0.708	0.869	0.832	0.400	0.869	0.869	0.436	0.869	0.840	0.544	0.902	0.533	0.689	0.934	0.198
	L32	0.847	0.376	0.847	0.817	0.371	0.847	0.847	0.372	0.847	0.831	0.537	0.885	0.553	0.702	0.923	0.260

Considering each metric separately, the best results in terms of maximum absolute value are **boldfaced**. Symbols M, m, and W illustrates macro, micro, and weighted, respectively

ited in samples due to the sensitivity and rarity of such data. Taking that into account, we implemented some techniques in the data preprocessing when creating the training pipeline for the investigated models.

The main technique adopted in the scope of training data preparation was the data augmentation. The term “data augmentation” allude to procedures for building iterative sampling and optimization algorithms via the incorporation of unobserved data or latent variables. Its main goal is to increase the volume, quality and diversity of training data. In terms of the distribution of the target dataset, the greater the size, diversity, and representativeness of the training data, the more effectively the DL model performs on unseen data. It introduces variability into the training data, which helps prevent the model from memorizing specific features of the training set and overfitting. Augmentation techniques such as rotation, flipping, and changes in brightness or contrast help the model become more robust to these variations, ensuring accurate identification across different scenarios. On one hand, the variety of training samples should be ample enough for the model to manage various deviations in image appearance and even noisy target instances. On the other hand, this encourages the model to learn more generalizable features that are applicable to a wider range of forensic cases.

In the context of this work, data augmentation helps in artificially increasing the size of the dataset, which is crucial for training deep learning models effectively. These techniques can help to balance the dataset by creating synthetic examples of the minority classes, improving the model’s ability to recognize and classify all types of wounds and distances. However, there is a wide variety of augmentation techniques available in the literature [23], and choosing the appropriate set of augmentations for the problem can be quite challenging. Specifically, we use 13 augmentation techniques, including variants of rotation, flipping, cropping and resizing, brightness and contrast adjustments, Gaussian noise, color jitter (changes in hue and saturation), elastic transfor-

mations, and grayscale conversion. Examples results of these augmentation techniques are depicted in Fig. 2.

The augmentation illustrated in Fig. 2 were chosen to handle variations that may occur in real-world forensic cases. Rotation, flipping, and further geometric transformations such as SR can help the model learn to recognize wounds from different angles, making the model more adaptable to variations in the size and anatomical location of wounds. Elastic and affine transformations can mimic the natural variability in skin and tissue appearance, making the model more robust to such changes. Varying the brightness and contrast can help the model become robust to different lighting conditions that may be encountered in real-world forensic images taken from different crime scenes. Some noise and color augmentations such as ISO noise, RBC, RGBShift, and Gaussian noise can make the model more resilient to image quality variations, such as those caused by different imaging equipment or environmental conditions. These augmentation techniques were discovered with AutoAulbument [24], an AutoML tool that automatically searches for the best augmentation policies based on training data. Using the its classification model, AutoAlbument provides a complete ready-to-use configuration for the augmentation pipeline. After discovered these augmentation techniques set, they are then incorporated in the training procedure with the Albumentations framework [25].

In practice, when using an augmentation library, it is essential that the quality of the synthetic samples does not deteriorate to a point of impairing performance on normal wound images. To avoid it, we combine synthetic examples with the original wound images rather than using only synthetic ones. The inclusion of the original preprocessed wound image sample was performed by adding the identity function as property in the Albumentations pipeline, which returns the image itself. This approach preserves the actual characteristics of the original data, ensuring that the model learns from authentic features and patterns present in the dataset.

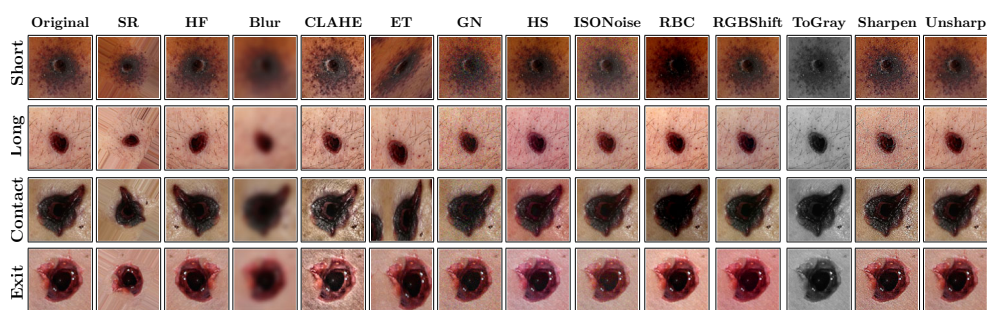


Fig. 2 Examples of gunshot wounds images and some their augmented versions. First line corresponds to a close range entry, second line corresponds to distance shot entry, third line presents the contact shot entry image, and the last line shows an example of exit gunshot wound. The results obtained from the used augmentation are depicted from left to

right: the original (preprocessed) image, Safe Rotation (SR), Horizontal Flip (HF), blur, Contrast-Limited Adaptive Histogram Equalization (CLAHE), Elastic Transform (ET), Gaussian Noise (GN), Hue Saturation (HS), ISO noise, Random Brightness Contrast (RBC), gray representation of image (ToGray), sharpen, and unsharp mask

By combining original and augmented images, the model learns to recognize patterns from both the natural data distribution and the variations introduced by augmentation. By using this combination of training data, the model receives balanced exposure to both real and augmented data, enhancing its robustness to variations in real-world scenarios.

After combining the training data, a data resampling strategy is implemented in order to further reduce the remaining imbalance after data augmentation. Particularly, a combination of over- and under-sampling using Synthetic Minority Over-sampling Technique (SMOTE) and Edited Nearest Neighbours. (ENN) is applied for handling class imbalance with fewer noisy instances, leading to improved performance and robustness of ML models. It combines the oversampling of the minority class using SMOTE with the cleaning of the dataset using ENN [26]. This resampling strategy is implemented via Imbalanced-learn [27], an open-source Python library designed to handle imbalanced datasets in ML. Finally, the class weights are computed to train the TorchVision models.

Experimental setup

The experiments were performed on a Linux machine (Ubuntu 22.04 LTS) using an AI software system developed by ourselves using the Python language (version 3.10). Computer hardware specifications included an Intel i7-8700 CPU, 32GB of RAM, and an RTX 3090 GPU. The TorchVision models are trained and tested separately. For each model and training round, 1000 epochs are used. The models trained with a batch size of 16 include RegnetY128GF and ViTH14. The models trained with a batch size of 32 include EfficientnetB6, EfficientnetB7, EfficientnetV2L, RegnetY128GF, Squeezenet10, SwinTransformerT, SwinTransformerS, SwinTransformerB, SwinTransformerV2T, SwinTransformerV2S, SwinTransformerV2B, ShuffleNetV2x20, ShuffleNetV2x15, ViTB16, ViTB32, ViTL16, ViTL32, ViTH14, and MaxVit. The remaining models not mentioned were trained with a batch size of 64. Early stopping with checkpointing is adopted to save the weights that perform best over the validation set during training.

Two cross-validation approaches were used: holdout and k-fold. For both approaches, the protocol consists of splitting the database into two content-independent subsets - one subset for training and another for testing. To avoid overfitting, the training and test data are split considering stratification based on the case number. This means that images associated with cases used in the training subset are not used for testing, and vice versa. With this constraint, 80% of cases are randomly selected for training, and the remaining 20% are used for testing in holdout cross-validation. Similarly, stratified grouped k-fold is adopted to ensure that each fold of the cross-validation procedure contains a balanced representa-

tion of these groups, providing a more realistic evaluation of the model's generalization capability. In this case, the groups used for stratification correspond to the crime cases.

We evaluate the performance of the tested models using regular classification metrics. More precisely, the models were assessed according to the following metrics:

- **Accuracy (ACC):** calculate the percentage of its correct predictions. It is defined as follows

$$\text{Accuracy} = \frac{TP + TN}{TP + FP + TN + FN}.$$

- **Precision:** because the accuracy score may have as drawbacks the imbalance problem and being uninformative as a standalone classification metric, the precision measures the ability of a classifier not to label as Positive a Negative sample. It is defined as follows

$$\text{Precision} = \frac{TP}{TP + FP}.$$

- **Recall:** it is a ratio of predictions of the Positive class that are Positive by ground truth to the total number of Positive samples. In other words, recall measures the ability of a classifier to detect Positive samples. It is defined as follows

$$\text{Recall} = \frac{TP}{TP + FN}.$$

- **F1-Score:** it is the harmonic mean of precision and recall, representing both metrics in one score. The highest possible value of an F-score is 1, indicating perfect precision and recall, and the lowest possible value is 0, if either precision or recall is zero. It is defined as follows:

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

- **Specificity:** it measures the ability of a classifier to correctly identify negative samples. It is calculated as the proportion of true negatives (TN) out of all actual negatives (TN + FP). Specificity is particularly useful when the cost of false positives is high, providing a comprehensive understanding of the performance of a classification model. It is defined as follows:

$$\text{Specificity} = \frac{TN}{TN + FP}$$

- **Area under the Curve (AUC):** it evaluates the performance of a classification model by measuring the area under the Receiver Operating Characteristic (ROC) curve, which plots the True Positive Rate (sensitivity)

against the False Positive Rate (1 - specificity) at various threshold settings. The ROC curve is created by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The AUC is then computed as the area under this ROC curve:

$$\begin{aligned} \text{AUC} &= \int_0^1 \text{TPR} d(\text{FPR}) \\ &= \sum_{i=1}^n \left(\frac{\text{TPR}_i + \text{FPR}_i}{2} \right) \cdot (\text{FPR}_i - \text{FPR}_{i-1}), \end{aligned}$$

where

$$\text{TPR} = \frac{TP}{TP + FN} \quad \text{and} \quad \text{FPR} = \frac{FP}{FP + TN},$$

are the ratio of true positives to all actual positives (also known as sensitivity) and the ratio of false positives to all actual negatives, respectively. Using the trapezoidal rule, n is the number of thresholds.

In the context of evaluating classification models, the above-described performance metrics can be aggregated with macro, micro, and weighted averages. Here is a brief overview of each:

- **Macro average** calculates the metric (e.g., precision, recall, or F1-Score) for each class individually and then takes the average of these metrics. In other words, it treats all classes equally, regardless of their frequency as follows:

$$\text{Macro } \Phi = \frac{1}{C} \sum_{i=1}^C \Phi_i,$$

where C is the number of classes and Φ_i is the performance metric for class i .

- **Micro average** aggregates the contributions of all classes to compute the average metric. It treats each instance equally, regardless of class.
- **Weighted average** calculates the metric for each class and then takes the average weighted by the number of instances in each class. It can be expressed as:

$$\text{Weighted } \Phi = \frac{\sum_{i=1}^C (\Phi_i \times \text{Number of Instances in Class}_i)}{\sum_{i=1}^C \text{Number of Instances in Class}_i}.$$

Cross-validation

In order to observe whether the ResNet152 results are stable, we conducted a K-fold cross-validation to ensure that this model is evaluated comprehensively and reliably. Using a

boxplot to illustrate the k-fold cross-validation results is an effective way to visualize the distribution of the performance metrics across the different folds. The boxplot displays the median, quartiles, and potential outliers of the wound images, providing a clear summary of the results.

Results

In this study, 2,551 wound images were analyzed, primarily from male subjects (95.09%)(Table 1), with most images showing wounds located on the trunk (51.35%)(Table 3). The majority of the wounds were classified as entry wounds (1,883), with 668 exit wounds. In terms of MLSD, most entry wounds were classified as distant (85.45%), while close-range and contact wounds were less frequent (Table 2). The age distribution of the cases indicated a predominance of younger victims, with a mean age of approximately 27 years and a significant skew towards individuals in their 20s (Fig. 3a and b). These findings highlight key demographic and injury pattern trends in gunshot wound cases.

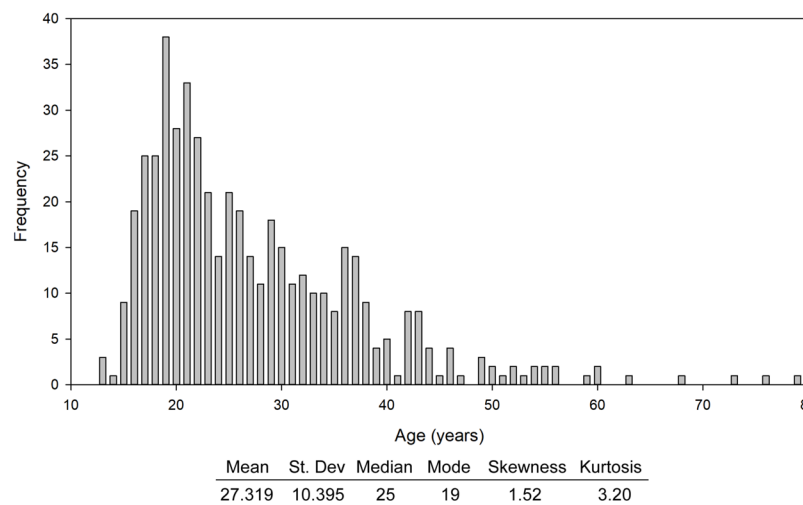
Wound type classification

Table 4 shows the accuracy, precision, recall and F1-score, AUC and specificity of wound type classification using a holdout approach. This table discriminates the performance in terms of neural network architectures and its variants provided in the TorchVision library. In this table, the boldfaced values highlight the best model. Moreover, Table 4 shows the micro, macro, and weighted averages for each metric. From these results, we can notice that the results vary a lot for different models. However, it is noticeable that ResNet presents the best results. Among the variants of ResNet, the ResNet152 achieves the best performance for all metrics. Since The ResNet 152 achieved the highest values for 14 out of 16 evaluated metrics, we chose this model as the most suitable for wound type classification.

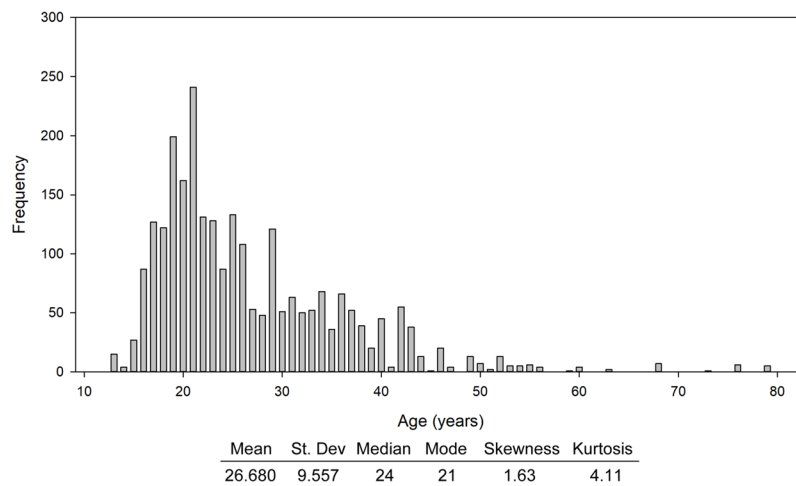
Figure 4a presents a boxplot with the results of the ResNet 152's wound type classification metrics.

Medico-legal shooting distance classification

Similarly to what was done for the classification of gunshot wound types, the results for the detection of MLSD are presented in Table 5. The ResNet 152 achieved the highest scores for the used metrics. The only exceptions were the macro precision values, which were higher for the ResNet 50 version, and the weighted specificity indices, for which the SqueezeNet 10 version reached the highest mark. Figure 4b presents the distribution of ResNet 152 MLSD classification metrics.



(a) Selected cases.



(b) Selected images.

Fig. 3 Frequency of age at time of death

Discussion

This study marks a significant advancement in forensic science as it is among the first to employ DL tools to classify gunshot wound patterns with extensive labels such as entry and exit wounds and MLSD, utilizing a broad array of neural network architectures. Unique in its approach, this research utilizes a considerable dataset of forensic images captured during real crime scene investigations, a feat not matched by any previous work. The implementation of these methodologies holds the potential to notably accelerate the pace of real criminal investigations by offering rapid, reliable analyses that enhance traditional forensic methods. Furthermore, the technology serves as a complementary tool for forensic experts during the crucial phases of examination and report drafting, promoting more accurate and informed decision-making.

Clearly, for ethical reasons, conducting research with a completely controlled sample of gunshot wounds on human cadavers, like studies of Phelps in 1897 [43], is virtually impractical. Therefore, many studies still use animal carcasses [17, 44, 45]. However, as much as these may approximate what is seen in humans, there are significant differences [46] that could lead to the incorrect training of the ML model for future use in human cases.

In this study, for classifying entry and exit gunshot wounds images, the use of ResNet152 achieved percentages, with the exception of the AUC results, closely align with those by Cheng et al. [16], which reported an accuracy of 87.99%, precision of 83.99%, recall of 87.71%, specificity of 88.19%, F1-score of 85.81% and AUC of 94.6% with a larger sample, comprising 2,028 entry and 1,314 exit wound images. However, the images do not appear to have been obtained at crime scenes, but rather during necropsy examinations,

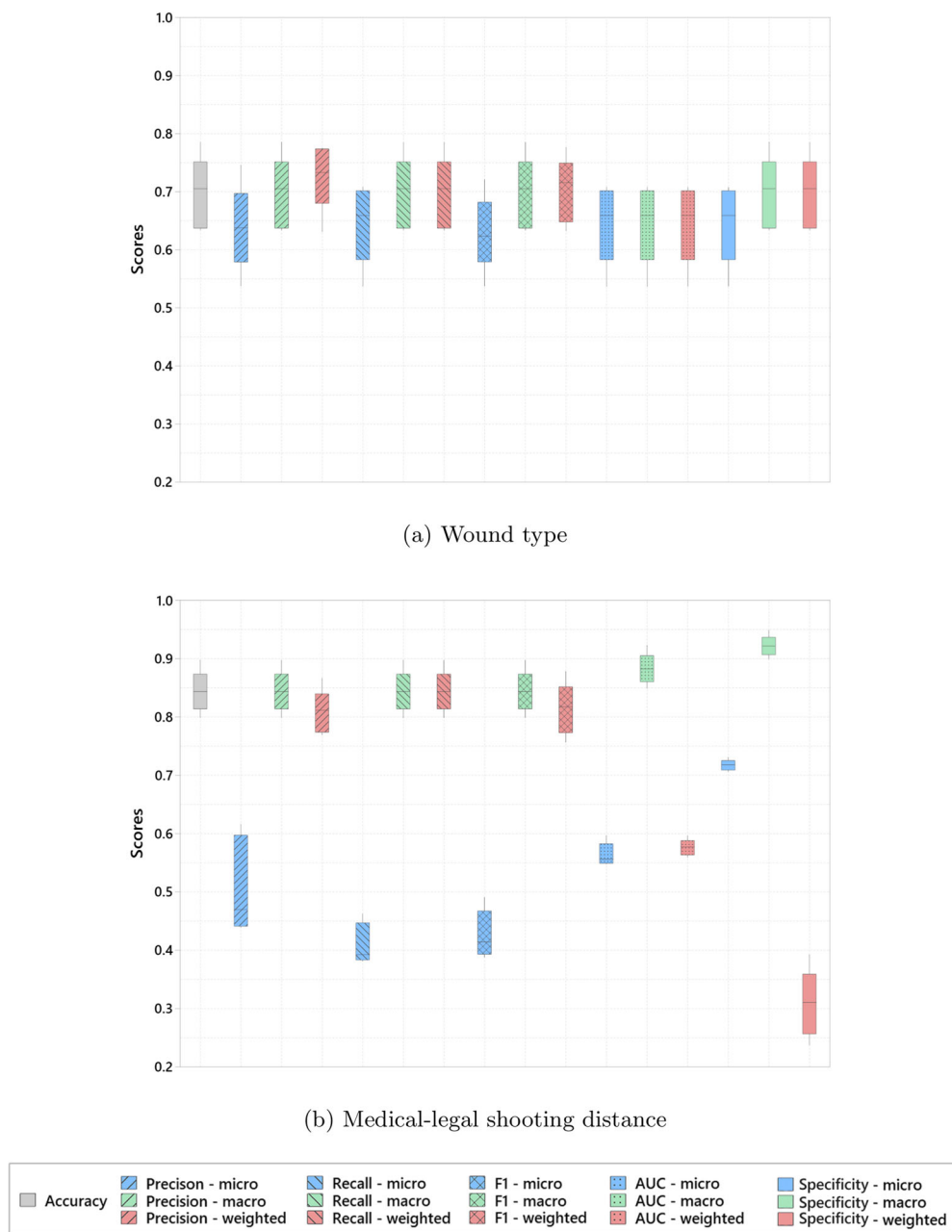


Fig. 4 Classification performance for wound type (a) and shooting distance (b) using ResNet 152 over the k-folds

which indicates a certain level of control over the captured images.

Additionally, in shooting distance classification, this study achieved a 92.48% accuracy rate using ResNet152. However, despite the high accuracy, the sample imbalance may have negatively impacted other metrics, including recall and the F1-Score. While newer models like ViT and EfficientNet offer advanced techniques and may perform exceptionally well in many scenarios, ResNet's combination of residual connections, depth, pretraining, and optimization make it a robust choice for a wide range of tasks, including the MLSD

classification task. In contrast, Oura et al. [17] reached a 98% accuracy rate. However, this higher accuracy was probably facilitated by their use of a small, controlled sample comprising only 204 images in a laboratory setting. It is important to note that their study utilized pig carcasses instead of wound on humans corpses, which could affect the applicability of their findings to real-world forensic scenarios.

With this, it becomes evident that one of the greatest challenges in developing ML models for image categorization is acquiring a database with high-quality scientific data [47]. Due to the sensitive content nature of the area, this chal-

lenge may be even greater in the forensic field. Therefore, the photographs used in this study were taken by different individuals, under various environmental and exposure conditions, with a range of cameras, and in the routine circumstances of forensic work at crime scenes, where standardization becomes virtually unfeasible. Given this, it is understood that a standardized database would not only misalign with the realities of forensic work, where numerous factors can alter the characteristics of injuries, but also require an exhaustive effort.

The challenge in standardizing a database of gunshot wound images is also evident in the discrepancy between the number of entry (1,883) and exit wounds (668). Logically, for an exit wound to occur, there must inevitably be an entry wound. Additionally, in this study, inclusion criterion #5 required that bullets be lodged in the body. Therefore, it is expected that the number of entry wounds would surpass that of exit wounds. Furthermore, even with access to a database containing millions of images from crime scene investigations, the utility of this dataset was significantly limited. Many images were related to other types of police incidents or featured different instruments (e.g., knives), had poor image quality, and faced restrictive criteria, all of which significantly constrained the acquisition of a larger, more useful sample. Despite these limitations, the sample used in this study more accurately reflects the conditions forensic teams encounter in real-world scenarios and deviates from the laboratory standardization typical of other research [16, 17]. This approach enhances the practical relevance of the findings to real-world forensic investigations.

In turn, the classification of distance shooting is used in cases where the characteristics of close-range and contact shots are not present (examples: smoke pattern and tattooing) [48] and typically occur at a distance range starting from 18 inches–2 feet (0.45 to 0.60 meters) [5, 49]. Therefore, for MLSD classification purposes, there is a tendency for the majority of injuries to be of the distance type, as per the sample obtained (85.45%). Unfortunately, for these reasons, it was not possible to achieve a more uniform sample regarding the number of images available in each distance category.

Depending on the location of the wound on the body, the characteristics displayed also change. The flaccidity of the region, as well as its elasticity and density, are some of the factors that influence the appearance of the wounds [50–52]. In the sample used, most of the wounds were located on the trunk (51.35%). However, even within the torso, there are quite distinct areas, such as the abdomen compared to the sternum area. In turn, given that the wounds analyzed are situated on the skin, age greatly affects the activity of collagen and, consequently, elasticity [53]. Since 66.67% of the cases and 70.29% of the images are from individuals below

thirty years of age, it is expected that the effect of age on elasticity did not have a significant influence. Finally, the differences in wounds due to sex, because of the arrangement of Langer's lines, skin thickness, and fat accumulation locations [52], were minimized by having a sample predominantly composed of male victims (95.09%).

In the field of scientific inquiry, it is crucial to acknowledge that AI serves not as a replacement for human expertise but as a formidable ally. The intrinsic value of human judgment and knowledge proves irreplaceable, especially in interpreting complex contexts and making well-informed decisions. Therefore, it is imperative to employ AI systems within forensic sciences as supportive instruments that enhance, rather than replace, the capabilities of forensic professionals [10, 47]. In real investigative scenarios, we can envision various applications, such as assisting pathologists in decision-making and accelerating criminal investigations. This approach ensures that the integration of AI into forensic practices augments the efficacy and accuracy of investigations, maintaining the indispensable balance between technological innovation and the nuanced understanding that only human experts can provide [47].

Limitations

This study presents significant strides in computer vision for the classification of gunshot wounds. However, while the dataset comprises images from actual forensic cases, there is a significant imbalance in the number of images available for each category of wounds, which could skew the accuracy of the model. Additionally, these images reflect a wide range of environmental and exposure conditions, contributing to variability in image quality that can affect the performance of the machine learning models. The imbalance in wound categories introduces potential selection bias, as certain types of wounds may be overrepresented, leading to models that perform well on the training data but struggle with generalization. Furthermore, the study lacks external validation, as the models were primarily validated using a dataset from a single institution located at Federal District, Brazil. Future work should focus on testing the models across diverse populations to improve robustness. To enhance model performance further, exploring resampling and training strategies, such as Data Center Interpolation with Improved Sparrow Search Algorithm [54], as well as incorporating classical feature-engineered models like Hu Moments, Haralick texture features, and local binary patterns, could offer complementary approaches for improving the robustness and accuracy of the models under real-world forensic conditions.

Conclusions

This study explored the use of fifty-nine deep neural network architectures for classifying gunshot wound images from real crime cases, focusing on distinguishing between entry and exit wounds, as well as determining the MLSD in a forensic context. The ResNet152 architecture demonstrated superior performance in wound type classification, achieving up to 86.90% accuracy, precision, recall, F1-score and specificity and an AUC of 82.09%. For MLSD, the same architecture reached an accuracy of 92.48%, precision, recall and F1-score up to 92.48%, AUC up to 94.36% and specificity up to 96.24%, although sample imbalance affected the metrics. Our findings underscore the challenges of standardizing wound images due to varying capture conditions, yet they reflect the practical realities of forensic work. This research highlights the significant potential of deep neural networks in forensic pathology, advocating for AI as a supportive tool to enhance, not replace, human expertise, ensuring the balance between technological innovation and expert judgment in forensic investigations.

Acknowledgements This work was supported by the Fundação de Apoio a Pesquisa do Distrito Federal (FAP-DF), by the Coordenação de Aperfeiçoamento de Pessoal de Nível Superior (CAPES), the Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq), and by the University of Brasília (UnB).

Author Contributions Author 1 took a lead role, overseeing the conceptualization of the study, data curation, resource management, investigation, and the preparation of the original draft. Authors 2, 3, and 4 were pivotal in data curation, ensuring that the necessary data for the study's objectives were accurately collected and organized. Author 5 contributed significantly to the study's conceptualization, developed the methodology, managed the software aspects, and participated in the writing and editing processes. Author 6 enhanced the research with software development, validation, and additional investigation. Authors 7 and 8 played crucial roles in conceptualizing the project, supervising the research activities, managing the project, and refining the manuscript through their critical review and editing, ensuring both the rigor and integrity of the study.

Funding Sources This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Data Availability The image dataset used in this paper is available at <https://github.com/pedrogarciafreitas/FDCPUnBGunshotDB.git> and the processed data used in the analysis were generated using the source code publicly available at <https://gitlab.com/lisa-unb/leguwoi.git>.

Declarations

The research involved the use of photographic records and reports from homicide and suicide cases handled by the PCDF in Brasília, Brazil. Proper authorization was obtained from the institution, and the study was approved by the Ethics Committee for Research at the University of Brasília under protocol number 54418221.9.0000.0030. As the study did not involve direct participation of individuals, consent to participate was not applicable.

Competing Interest The authors have no competing interests to declare.

References

1. UNODC: Global study on homicide 2023. <https://www.unodc.org/unodc/en/data-and-analysis/global-study-on-homicide.html>. Accessed 08 Mar 2024
2. Wisevoter: Gun deaths by country 2023. <https://wisevoter.com/country-rankings/gun-deaths-by-country>. Accessed 08 Mar 2024
3. Kleypas DA, Badiye A (2024) Evidence collection vol 2, pp 113–116. <https://pubmed.ncbi.nlm.nih.gov/28722882/>
4. Shrestha R, Kanchan T, Krishan K (2024) Gunshot wounds forensic pathology, pp 1–8. StatPearls Publishing, Treasure Island, Florida, United States. <http://www.ncbi.nlm.nih.gov/pubmed/32310579>
5. Denton JS, Segovia A, Filkins JA (2006) Practical pathology of gunshot wounds. Arch Pathol Lab Med 130(9):1283–9. <https://doi.org/10.5858/2006-130-1283-PPOGW>
6. Di Maio VJM (2015) Gunshot wounds: practical aspects of firearms, ballistics, and forensic techniques, 3rd edn. CRC Press, Boca Raton, Florida, United States, p 377
7. Saukko PJ, Knight B (2016) Knight's Forensic Pathology, 4th edn. CRC Press, Boca Raton, Florida, United States
8. Adelson L (1961) A microscopic study of dermal gunshot wounds. Am J Clin Pathol 35(5):393–402. <https://doi.org/10.1093/ajcp/35.5.393>
9. Dror IE, Pierce ML (2020) ISO standards addressing issues of bias and impartiality in forensic work. J Forensic Sci 65(3):800–808. <https://doi.org/10.1111/1556-4029.14265>
10. Vodanović M, Subašić M, Milošević D, Galić I, Brkić H (2023) Artificial intelligence in forensic medicine and forensic dentistry. J Forensic Odonto-Stomatol 41(2):30–41
11. Tournois L, Troussat V, Hatsch D, Delabarde T, Ludes B, Lefèvre T (2023) Artificial intelligence in the practice of forensic medicine: a scoping review. Int J Legal Med. <https://doi.org/10.1007/s00414-023-03140-9>
12. Lefèvre T, Tournois L (2023) Artificial intelligence and diagnostics in medicine and forensic science. Diagnostics 13(23):1–12. <https://doi.org/10.3390/diagnostics13233554>
13. Wankhade TD, Ingale SW, Mohite PM, Bankar NJ (2022) Artificial intelligence in forensic medicine and toxicology: the future of forensic medicine. Cureus 14(8). <https://doi.org/10.7759/cureus.28376>
14. Galante N, Cotroneo R, Furci D, Lodetti G, Casali MB (2022) Applications of artificial intelligence in forensic sciences: current potential benefits, limitations and perspectives. Int J Legal Med 445–458. <https://doi.org/10.1007/s00414-022-02928-5>
15. Margagliotti G, Bollé T (2019) Machine learning & forensic science. Forensic Sci Int 298:138–139. <https://doi.org/10.1016/j.foresciint.2019.02.045>
16. Cheng J, Schmidt C, Wilson A, Wang Z, Hao W, Pantanowitz J, Morris C, Tashjian R, Pantanowitz L (2024) Artificial intelligence for human gunshot wound classification. J Pathol Inform 15:100361. <https://doi.org/10.1016/j.jpi.2023.100361>
17. Oura P, Junno A, Junno J-A (2021) Deep learning in forensic gunshot wound interpretation—a proof-of-concept study. Int J Legal Med 135:2101–2106. <https://doi.org/10.1007/s00414-021-02566-3>
18. Molina DK, Dimaio V, Cave R (2013) Gunshot wounds: a review of firearm type, range, and location as pertaining to manner of death. Am J Forensic Med Pathol 34:366–371. <https://doi.org/10.1097/PAF.0000000000000065>
19. Marcel S, Rodriguez Y (2010) Torchvision the machine-vision package of torch. In: Bimbo AD, Chang S, Smeulders AWM (eds) Proceedings of the 18th international conference on multimedia

- 2010, Firenze, Italy, October 25–29, 2010, pp 1485–1488. Association for Computing Machinery, New York, New York, United States. <https://doi.org/10.1145/1873951.1874254>
20. Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, Killeen T, Lin Z, Gimelshein N, Antiga L, Desmaison A, Köpf A, Yang EZ, DeVito Z, Raison M, Tejani A, Chilamkurthy S, Steiner B, Fang L, Bai J, Chintala S (2019) Pytorch: an imperative style, high-performance deep learning library. In: Wallach HM, Larochelle H, Beygelzimer A, d'Alché-Buc F, Fox EB, Garnett R (eds) Advances in neural information processing systems 32: annual conference on neural information processing systems 2019, NeurIPS 2019, December 8–14, 2019, Vancouver, BC, Canada, pp 8024–8035. <https://proceedings.neurips.cc/paper/2019/hash/bdbca288fee7f92f2bfa9f7012727740-Abstract.html>
 21. Tietz M, Fan TJ, Nouri D, Bossan B (2017) skorch Developers: Skorch: a scikit-learn compatible neural network library that wraps PyTorch. <https://skorch.readthedocs.io/en/stable/>
 22. Buitinck L, Louppe G, Blondel M, Pedregosa F, Mueller A, Grisel O, Niculae V, Prettenhofer P, Gramfort A, Grobler J, Layton R, VanderPlas J, Joly A, Holt B, Varoquaux G (2013) Api design for machine learning software: experiences from the scikit-learn project, vol. abs/1309.0238. <http://dblp.uni-trier.de/db/journals/corr/corr1309.html#BuitinckLBPMGNPGLVJHV13>
 23. Mumuni A, Mumuni F (2022) Data augmentation: A comprehensive survey of modern approaches. *Array* 16:100258. <https://doi.org/10.1016/J.ARRAY.2022.100258>
 24. Hataya R, Zdenek J, Yoshizoe K, Nakayama H (2020) Faster autoaugment: learning augmentation strategies using backpropagation. In: Vedaldi A, Bischof H, Brox T, Frahm J (eds) Computer Vision - ECCV 2020 - 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part XXV. Lecture Notes in Computer Science, vol 12370, pp 1–16. Springer, Cham, Switzerland. https://doi.org/10.1007/978-3-030-58595-2_1
 25. Buslaev A, Iglovikov VI, Khvedchenya E, Parinov A, Druzhinin M, Kalinin AA (2020) Albumentations: fast and flexible image augmentations. *Information* 11(2):125. <https://doi.org/10.3390/INFO11020125>
 26. Batista GEAPA, Prati RC, Monard MC (2004) A study of the behavior of several methods for balancing machine learning training data. *SIGKDD Explor* 6(1):20–29. <https://doi.org/10.1145/1007730.1007735>
 27. Lemaître G, Nogueira F, Aridas CK (2017) Imbalanced-learn: a python toolbox to tackle the curse of imbalanced datasets in machine learning. *J Mach Learn Res* 18(17):1–5
 28. Krizhevsky A, Sutskever I, Hinton GE (2017) ImageNet classification with deep convolutional neural networks. *Commun ACM* 60(6):84–90. <https://doi.org/10.1145/3065386>
 29. Huang G, Liu Z, Maaten L, Weinberger KQ (2017) Densely connected convolutional networks, pp 2261–2269. <https://doi.org/10.1109/CVPR.2017.243>
 30. Tan M, Le QV (2019) Efficientnet: rethinking model scaling for convolutional neural networks, vol 97, pp 6105–6114
 31. Tan M, Le QV (2021) Efficientnetv2: smaller models and faster training, vol 139, pp 10096–10106
 32. Tu Z, Talebi H, Zhang H, Yang F, Milanfar P, Bovik AC, Li Y (2022) Maxvit: multi-axis vision transformer, vol 13684, pp 459–479. https://doi.org/10.1007/978-3-031-20053-3_27
 33. Tan M, Chen B, Pang R, Vasudevan V, Sandler M, Howard A, Le QV (2019) Mnasnet: platform-aware neural architecture search for mobile, pp 2820–2828. <https://doi.org/10.1109/CVPR.2019.00293>
 34. Sandler M, Howard AG, Zhu M, Zhmoginov A, Chen L (2018) Mobilenetv2: inverted residuals and linear bottlenecks, pp 4510–4520. <https://doi.org/10.1109/CVPR.2018.00474>
 35. Howard A, Pang R, Adam H, Le QV, Sandler M, Chen B, Wang W, Chen L, Tan M, Chu G, Vasudevan V, Zhu Y (2019) Searching for mobilenetv3, pp 1314–1324. <https://doi.org/10.1109/ICCV.2019.00140>
 36. Radosavovic I, Kosaraju RP, Girshick RB, He K, Dollár P (2020) Designing network design spaces, pp 10425–10433. <https://doi.org/10.1109/CVPR42600.2020.01044>
 37. He K, Zhang X, Ren S, Sun J (2016) Deep residual learning for image recognition, pp 770–778. <https://doi.org/10.1109/CVPR.2016.90>
 38. Ma N, Zhang X, Zheng H, Sun J (2018) Shufflenet V2: practical guidelines for efficient CNN architecture design, vol 11218, pp 122–138. https://doi.org/10.1007/978-3-030-01264-9_8
 39. Iandola FN, Moskewicz MW, Ashraf K, Han S, Dally WJ, Keutzer K (2016) Squeezenet: Alexnet-level accuracy with 50x fewer parameters and <1mb model size. *CoRR arXiv:1602.07360*
 40. Liu Z, Lin Y, Cao Y, Hu H, Wei Y, Zhang Z, Lin S, Guo B (2021) Swin transformer: hierarchical vision transformer using shifted windows, pp 9992–10002. <https://doi.org/10.1109/ICCV48922.2021.00986>
 41. Simonyan K, Zisserman A (2015) Very deep convolutional networks for large-scale image recognition. In: Bengio Y, LeCun Y (eds) 3rd International Conference on Learning Representations, ICLR 2015, San Diego, CA, USA, May 7–9, 2015, Conference Track Proceedings. *arXiv:1409.1556*
 42. Dosovitskiy A, Beyer L, Kolesnikov A, Weissenborn D, Zhai X, Unterthiner T, Dehghani M, Minderer M, Heigold G, Gelly S, Uszkoreit J, Houlsby N (2021) An image is worth 16x16 words: transformers for image recognition at scale. In: International conference on learning representations. <https://openreview.net/forum?id=YicbFdNTTy>
 43. Phelps C (1897) Traumatic injuries of the brain and its membranes. Appleton, New York, New York, United States
 44. Pircher R, Preiß D, Pollak S, Thierauf-Emberger A, Perdekamp MG, Geisenberger D (2017) The influence of the bullet shape on the width of abrasion collars and the size of gunshot entrance holes. *Int J Legal Med* 131(2):441–445. <https://doi.org/10.1007/s00414-016-1501-6>
 45. Rainio J, Lalu K, Ranta H, Penttilä A (2003) Morphology of experimental assault rifle skin wounds. *Int J Legal Med* 117(1):19–26. <https://doi.org/10.1007/s00414-002-0308-9>
 46. Bartell TH, Mustoe TA (1989) Animal models of human tissue expansion. *Plast Reconstr Surg* 83(4):681–6. <https://doi.org/10.1097/00006534-198904000-00014>
 47. Piraianu AI, Fulga A, Musat CL, Ciobotaru OR, Poalelungi DG, Stamate E, Ciobotaru O, Fulga I (2023) Enhancing the evidence with algorithms: how artificial intelligence is transforming forensic medicine. *Diagnostics* 13(18). <https://doi.org/10.3390/diagnostics13182992>
 48. Smith JC (1964) Investigation of fatal gunshot wounds. *Cleveland-Marshall Law Rev* 13(2):296–308
 49. Svensson A, Wendel O, Nicol JD (1965) Techniques of crime scene investigation. American Elsevier Publishing Company, New York, New York, United States
 50. Mahoney P, Carr D, Arm R, Gibb I, Hunt N, Delaney RJ (2017) Ballistic impacts on an anatomically correct synthetic skull with a surrogate skin/soft tissue layer. *Int J Legal Med* 1–12. <https://doi.org/10.1007/s00414-017-1737-9>
 51. Humphrey C, Kumaratilake J (2016) Ballistics and anatomical modelling – a review. *Legal Med* 23:21–29. <https://doi.org/10.1016/j.legalmed.2016.09.002>
 52. Ridge MD, Wright V (1966) The directional effects of skin. A bio-engineering study of skin with particular reference to Langer's lines. *J Invest Dermatol* 46(4):341–346. <https://doi.org/10.1038/jid.1966.54>
 53. Maiden NR, Byard RW (2016) Unpredictable tensile strength biomechanics may limit thawed cadaver use for simulant

research. *Aust J Forensic Sci* 48(1):54–58. <https://doi.org/10.1080/00450618.2015.1025842>

54. Gu Q, Tian J, Li X, Jiang S (2022) A novel random forest integrated model for imbalanced data classification problem. *Knowl-Based Syst* 250:109050. <https://doi.org/10.1016/j.knosys.2022.109050>

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Renato Queiroz Nogueira Lira¹  · **Luana Geovana Motta de Sousa¹** · **Maisa Luana Memoria Pinho¹** · **Renan Cesar Pinto da Silva Andrade de Lima¹** · **Pedro Garcia Freitas²**  · **Bruno Scholles Soares Dias³**  · **Andreia Cristina Breda de Souza⁴**  · **André Ferreira Leite¹** 

✉ Renato Queiroz Nogueira Lira
renato.lira@aluno.unb.br; renatolira@msn.com

Luana Geovana Motta de Sousa
luana.geovana@aluno.unb.br

Maisa Luana Memoria Pinho
maisa.pinho@aluno.unb.br

Renan Cesar Pinto da Silva Andrade de Lima
renan.lima@aluno.unb.br

Pedro Garcia Freitas
pedro.freitas@unb.br

Bruno Scholles Soares Dias
brunoscholles98@gmail.com

Andreia Cristina Breda de Souza
andreiacrisbreda@gmail.com

André Ferreira Leite
andreileite@unb.br

¹ Department of Dentistry, University of Brasília, Campus Darcy Ribeiro, Asa Norte, Brasília 70910-900, Distrito Federal, Brazil

² Department of Computer Science, University of Brasília, Campus Darcy Ribeiro, Asa Norte, Brasília 70910-900, Distrito Federal, Brazil

³ Department of Electrical Engineering, University of Brasília, Campus Darcy Ribeiro, Asa Norte, Brasília 70910-900, Distrito Federal, Brazil

⁴ Department of Forensic Dentistry and Public Health, Federal University of Rio de Janeiro, Cidade Universitária, Rio de Janeiro 21941-617, Rio de Janeiro, Brazil